

3.3 Work, energy and power

Words like *energy*, *power* and *work* have very precise meaning in physics. In this section the important link between work done and energy is explored. Learners have the opportunity to apply the important principle of conservation of energy to a range of situations. The

analysis of energy transfers provides the opportunity for calculations of efficiency and the subsequent evaluation of issues relating to the individual and society (HSW2, 5, 8, 9, 10, 11, 12).

3.3.1 Work and conservation of energy

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) work done by a force; the unit joule	HSW2
(b) $W = Fx \cos \theta$ for work done by a force	
(c) the principle of conservation of energy	
(d) energy in different forms; transfer and conservation	
(e) transfer of energy is equal to work done.	